

# Rajalakshmi Engineering College

Name: Harsan J  
Email: 240701179@rajalakshmi.edu.in  
Roll no: 240701179  
Phone: 8925466195  
Branch: REC  
Department: CSE - Section 10  
Batch: 2028  
Degree: B.E - CSE

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## 2024\_28\_III\_OOPS Using Java Lab

### REC\_2028\_OOPS using Java\_Week 11

Attempt : 1  
Total Mark : 20  
Marks Obtained : 10

#### Section 1 : Project

##### 1. Problem Statement

Create a JDBC-based Hospital Management System that handles runtime input to manage patient records. The system should allow users to:

Add a new patient (patient ID, name, age, status).

Update a patient's status.

View a specific patient's record by patient ID.

Display all patient records in the database.

Exit the application.

The system should connect to a MySQL database using the following default credentials:

DB URL: jdbc:mysql://localhost/ri\_db

USER: test

PWD: test123

The patients table has already been created with the following structure:

Table Name: patients

### ***Input Format***

The first line of input consists of an integer choice, representing the operation to be performed:

(1 for Add Patient, 2 for Update Patient Status, 3 for View Patient Record, 4 for Display All Patients, 5 for Exit)

For choice 1 (Add Patient):

- The second line consists of an integer patient\_id.
- The third line consists of a string name.
- The fourth line consists of an integer age.
- The fifth line consists of a string status.

For choice 2 (Update Patient Status):

- The second line consists of an integer patient\_id.
- The third line consists of a string new\_status.

For choice 3 (View Patient Record):

- The second line consists of an integer patient\_id.

For choice 4 (Display All Patients):

- No additional inputs are required.

For choice 5 (Exit):

- No additional inputs are required.

### **Output Format**

For choice 1 (Add Patient):

- Print "Patient added successfully" if the patient was added.
- Print "Failed to add patient." if the insertion failed.

For choice 2 (Update Patient Status):

- Print "Patient status updated successfully" if the update was successful.
- Print "Patient not found." if the specified patient ID does not exist.

For choice 3 (View Patient Record):

- Display the patient details in the format:
- ID: [patient\_id] | Name: [name] | Age: [age] | Status: [status]
- Print "Patient not found." if the specified patient ID does not exist.

For choice 4 (Display All Patients):

- Display each patient on a new line in the format:
- ID | Name | Age | Status
- If no records are available, print nothing (or handle it with an appropriate message if desired).

For choice 5 (Exit):

- Print "Exiting Hospital Management System."

For invalid input:

- Print "Invalid choice. Please try again."

### **Sample Test Case**

Input: 1

101

John Doe

45

Admitted

4

5

Output: Patient added successfully

ID | Name | Age | Status

101 | John Doe | 45 | Admitted

Exiting Hospital Management System.

### **Answer**

```
import java.sql.*;
```

```
import java.util.Scanner;
```

```
class HospitalManagementSystem {
```

```
    public static void main(String[] args) {
```

```
        try (Connection conn = DriverManager.getConnection("jdbc:mysql://localhost/ri_db", "test", "test123"));
```

```
            Scanner scanner = new Scanner(System.in)) {
```

```
                boolean running = true;
```

```
                while (running) {
```

```
                    int choice = scanner.nextInt();
```

```
                    switch (choice) {
```

```
                        case 1:
```

```
                            addPatient(conn, scanner);
```

```
                            break;
```

```
                        case 2:
```

```
                            updatePatientStatus(conn, scanner);
```

```
                            break;
```

```
                        case 3:
```

```
                            viewPatientRecord(conn, scanner);
```

```
                            break;
```

```
                        case 4:
```

```
                            displayAllPatients(conn);
```

```
                            break;
```

```
                        case 5:
```

```
                            System.out.println("Exiting Hospital Management System.");
```

```
                            running = false;
```

```
                            break;
```

```
                    default:
```

```
                        System.out.println("Invalid choice. Please try again.");
```

```
                }
```

```
    }  
  } catch (SQLException e) {  
    e.printStackTrace();  
  }  
}
```

```
public static void addStudent(Connection conn, Scanner scanner){
```

```
    int studentId = scanner.nextInt();  
    scanner.nextLine(); // Consume newline
```

```
    String name = scanner.nextLine();
```

```
    String gradeLevel = scanner.nextLine();
```

```
    double gpa = scanner.nextDouble();
```

```
    String insertQuery = "INSERT INTO students (student_id, name, grade_level,  
gpa) VALUES (?, ?, ?, ?)";
```

```
    try (PreparedStatement stmt = conn.prepareStatement(insertQuery))
```

```
{
```

```
    stmt.setInt(1, studentId);  
    stmt.setString(2, name);  
    stmt.setString(3, gradeLevel);  
    stmt.setDouble(4, gpa);
```

```
    int rowsInserted = stmt.executeUpdate();  
    System.out.println(rowsInserted > 0 ? "Student added successfully" :  
"Failed to add student.");
```

```
} catch (SQLException e)
```

```
{
```

```
    System.out.println("Error adding student: " + e.getMessage());
```

```
}
```

```
}
```

```
public static void updateGrades(Connection conn, Scanner scanner)
```

```
{
```

```
    int studentId = scanner.nextInt();
```

```
    double newGpa = scanner.nextDouble();
```

```
    // Validate GPA range
```

```
    if (newGpa < 0.0 || newGpa > 4.0)
```

```
{
```

```
    System.out.println("GPA must be between 0.0 and 4.0.");
```

```
    return;
```

```
}
```

```
String updateQuery = "UPDATE students SET gpa = ? WHERE student_id = ?";
```

```
try (PreparedStatement stmt = conn.prepareStatement(updateQuery))
```

```
{
```

```
    stmt.setDouble(1, newGpa);
```

```
    stmt.setInt(2, studentId);
```

```
    int rowsUpdated = stmt.executeUpdate();
```

```
    System.out.println(rowsUpdated > 0 ? "GPA updated successfully" :  
"Student not found.");
```

```
} catch (SQLException e)
```

```
{
```

```
    System.out.println("Error updating GPA: " + e.getMessage());
```

```
}
```

```
}
```

```
public static void viewStudentRecord(Connection conn, Scanner scanner)
```

```
{
```

```
    int studentId = scanner.nextInt();
```

```
    String selectQuery = "SELECT * FROM students WHERE student_id = ?";  
    try (PreparedStatement stmt = conn.prepareStatement(selectQuery))
```

```
{
```

```
        stmt.setInt(1, studentId);
```

```
        ResultSet rs = stmt.executeQuery();  
        if (rs.next())
```

```
{
```

```
            System.out.printf("ID: %d | Name: %s | Grade Level: %s | GPA: %.2f%n",  
                               rs.getInt("student_id"),  
                               rs.getString("name"),  
                               rs.getString("grade_level"),  
                               rs.getDouble("gpa"));
```

```
        } else
```

```
{  
    System.out.println("Student not found.");  
  
}
```

```
} catch (SQLException e)
```

```
{  
    System.out.println("Error retrieving student record: " + e.getMessage());  
  
}
```

```
}  
  
public static void displayAllStudents(Connection conn)
```

```
{  
    String displayQuery = "SELECT * FROM students";  
    try (Statement stmt = conn.createStatement();  
        ResultSet rs = stmt.executeQuery(displayQuery))  
  
    {
```

```
  
        System.out.println("ID | Name | Grade Level | GPA");  
        while (rs.next())  
  
        {
```



```
        System.out.printf("%d | %s | %s | %.2f%n",
            rs.getInt("student_id"),
            rs.getString("name"),
            rs.getString("grade_level"),
            rs.getDouble("gpa"));
    }

} catch (SQLException e)
{
    System.out.println("Error displaying students: " + e.getMessage());
}

}

}
```

**Status : Wrong**

**Marks : 0/10**

## 2. Problem Statement

In ABC Corporation, employee records are stored in a database.

To efficiently manage employee details using Java and JDBC, you are tasked with building an Employee Management System that supports the following functionalities:

Adding a new employee

Updating an employee's salary

Viewing an employee's details

Displaying all employees

You are given two files:

File 1: Employee.java (POJO Class)

This class represents the Employee entity.

An Employee contains the following details:

Field Description

employeeId Unique Employee ID (Integer)

name Employee Name (String)

department Employee Department (String)

salary Employee Salary (Double)

Students must write code in the marked area:

```
class Employee {  
    private int employeeId;  
    private String name;  
    private String department;  
    private double salary;  
    public Employee() {}  
    public Employee(int employeeId, String name, String department, double salary) {  
        // write your code here  
    }  
    // Include getters and setters  
}
```

Expected in this part:

Assign parameter values to instance variables inside the constructor.

Add getters and setters for all attributes.

File 2: EmployeeDAO.java (Data Access Layer)

This class handles all database operations using JDBC.

Students must complete the missing JDBC logic in the following methods:

```
class EmployeeDAO {

    public void addEmployee(Connection conn, Employee employee) throws
    SQLException {
        // write your code here
    }

    public void updateSalary(Connection conn, int employeeId, double
    newSalary) throws SQLException {
        // write your code here
    }

    public void deleteEmployee(Connection conn, int employeeId) throws
    SQLException {
        // write your code here
    }

    public Employee viewEmployeeRecord(Connection conn, int employeeId)
    throws SQLException {
        // write your code here
    }

    public List<Employee> displayAllEmployees(Connection conn) throws
    SQLException {
        // write your code here
    }
}
```

```

    }
    private Employee mapToEmployee(ResultSet rs) throws SQLException {
        return new Employee(
            // write your code here
        );
    }
}

```

Expected in this part:

Write SQL queries for INSERT, UPDATE, DELETE, SELECT.

Execute queries using PreparedStatement or Statement.

Map ResultSet rows to Employee objects using mapToEmployee().

Return a List<Employee> where required.

The system should connect to a MySQL database using the following default credentials:

DB URL: jdbc:mysql://localhost/ri\_dbUsername: testPassword: test123

The employees table has already been created with the following structure:

### ***Input Format***

The first line of input consists of an integer choice, representing the operation to be performed:

(1 for Add Employee, 2 for Update Salary, 3 for View Employee Record, 4 for Display All Employees, 5 for Exit)

For choice 1 (Add Employee):

1. The second line consists of an integer employee\_id.
2. The third line consists of a string name.
3. The fourth line consists of a string department.

4. The fifth line consists of a double salary (must be at least 30000).

For choice 2 (Update Salary):

1. The second line consists of an integer employee\_id.
2. The third line consists of a double new\_salary (must be at least 30000).

For choice 3 (View Employee Record):

1. The second line consists of an integer employee\_id.

For choice 4 (Display All Employees).

For choice 5 (Exit).

### **Output Format**

For choice 1 (Add Employee),

1. Print "Employee added successfully" if the employee was added.

For choice 2 (Update Salary),

1. Print "Salary updated successfully" if the salary update was successful.
2. Print "Employee not found." if the specified employee ID does not exist.
3. Print "Salary must be at least 30000." if the provided salary is below the minimum.

For choice 3 (View Employee Record),

1. Display the employee details in the format:
2. ID: [employee\_id] | Name: [name] | Department: [department] | Salary: [salary]
3. Print "Employee not found." if the specified employee ID does not exist.

For choice 4 (Display All Employees),

1. Display each employee on a new line in the format:
2. ID | Name | Department | Salary

For choice 5 (Exit),

1. Print "Exiting Employee Management System."

For invalid input:

1. Print "Invalid choice. Please try again."

### **Sample Test Case**

Input: 1

101

Alice Johnson

Engineering

31000.75

4

6

5

Output: Employee added successfully

ID | Name | Department | Salary

101 | Alice Johnson | Engineering | 31000.75

Invalid choice. Please try again.

Exiting Employee Management System.

### **Answer**

```
import java.sql.*;
```

```
import java.util.Scanner;
```

```
class Employee
```

```
{
```

```
private int employeeId;
private String name;
private String department;
private double salary;

// Constructor
public Employee(int employeeId, String name, String department, double
salary)

{

    this.employeeId = employeeId;
    this.name = name;
    this.department = department;
    this.salary = salary;

}

// Getters and Setters
public int getEmployeeId()

{

    return employeeId;
}

public void setEmployeeId(int employeeId)

{

    this.employeeId = employeeId;
}

public String getName()

{

    return name;
```

```
}  
    public void setName(String name)
```

```
{  
    this.name = name;
```

```
}
```

```
    public String getDepartment()
```

```
{  
    return department;
```

```
}
```

```
    public void setDepartment(String department)
```

```
{
```

```
    this.department = department;
```

```
}
```

```
    public double getSalary()
```

```
{
```

```
    return salary;
```

```
}
```

```
    public void setSalary(double salary)
```

```
{
```

```
    this.salary = salary;
```

```
}
```

```
}
```



```
class EmployeeManagementSystem
```

```
{
```

```
    // Add Employee
```

```
    public static void addEmployee(Connection conn, Scanner scanner)
```

```
    {
```

```
        int employeeId = scanner.nextInt();
```

```
        scanner.nextLine(); // Consume newline
```

```
        String name = scanner.nextLine();
```

```
        String department = scanner.nextLine();
```

```
        double salary = scanner.nextDouble();
```

```
        if (salary < 30000)
```

```
        {
```

```
            System.out.println("Salary must be at least 30000.");
```

```
            return;
```

```
        }
```

```
        // Create an Employee POJO object
```

```
        Employee employee = new Employee(employeeId, name, department,  
salary);
```

```
        String insertQuery = "INSERT INTO employees (employee_id, name,  
department, salary) VALUES (?, ?, ?, ?)";
```

```
        try (PreparedStatement stmt = conn.prepareStatement(insertQuery))
```

```
        {
```

```
            stmt.setInt(1, employee.getEmployeeId());
```

```
stmt.setString(2, employee.getName());
stmt.setString(3, employee.getDepartment());
stmt.setDouble(4, employee.getSalary());

int rowsInserted = stmt.executeUpdate();
System.out.println(rowsInserted > 0 ? "Employee added successfully" :
"Failed to add employee.");
```

```
} catch (SQLException e)
```

```
{
```

```
System.out.println("Error adding employee: " + e.getMessage());
```

```
}
```

```
}
```

```
// Update Salary
```

```
public static void updateSalary(Connection conn, Scanner scanner)
```

```
{
```

```
int employeeId = scanner.nextInt();
double newSalary = scanner.nextDouble();
```

```
if (newSalary < 30000)
```

```
{
```

```
System.out.println("Salary must be at least 30000.");
return;
```

```
}
```

```
String updateQuery = "UPDATE employees SET salary = ? WHERE  
employee_id = ?";  
try (PreparedStatement stmt = conn.prepareStatement(updateQuery))
```

```
{
```

```
    stmt.setDouble(1, newSalary);  
    stmt.setInt(2, employeeId);
```

```
    int rowsUpdated = stmt.executeUpdate();  
    System.out.println(rowsUpdated > 0 ? "Salary updated successfully" :  
"Employee not found.");
```

```
} catch (SQLException e)
```

```
{
```

```
    System.out.println("Error updating salary: " + e.getMessage());
```

```
}
```

```
}
```

```
// View Employee Record
```

```
public static void viewEmployeeRecord(Connection conn, Scanner scanner)
```

```
{
```

```
    int employeeId = scanner.nextInt();  
    String selectQuery = "SELECT * FROM employees WHERE employee_id = ?";
```

```
    try (PreparedStatement stmt = conn.prepareStatement(selectQuery))
```

```
{
```

```
stmt.setInt(1, employeeId);
ResultSet rs = stmt.executeQuery();
```

```
if (rs.next())
```

```
{
```

```
    Employee employee = new Employee(
        rs.getInt("employee_id"),
        rs.getString("name"),
        rs.getString("department"),
        rs.getDouble("salary")
    );
```

```
    System.out.printf("ID: %d | Name: %s | Department: %s | Salary: %.2f%n",
        employee.getEmployeeId(),
        employee.getName(),
        employee.getDepartment(),
        employee.getSalary());
```

```
} else
```

```
{
```

```
    System.out.println("Employee not found.");
```

```
}
```

```
} catch (SQLException e)
```

```
{
```

```
    System.out.println("Error retrieving employee record: " + e.getMessage());
```

```
}
```

```
}
```

```
// Display All Employees
```

```
public static void displayAllEmployees(Connection conn)
```

```
{
```

```
    String displayQuery = "SELECT * FROM employees";
```

```
    try (Statement stmt = conn.createStatement();
```

```
        ResultSet rs = stmt.executeQuery(displayQuery))
```

```
    {
```

```
        System.out.println("ID | Name | Department | Salary");
```

```
        while (rs.next())
```

```
        {
```

```
            Employee employee = new Employee(
```

```
                rs.getInt("employee_id"),
```

```
                rs.getString("name"),
```

```
                rs.getString("department"),
```

```
                rs.getDouble("salary")
```

```
            );
```

```
            System.out.printf("%d | %s | %s | %.2f%n",
```

```
                employee.getEmployeeId(),
```

```
                employee.getName(),
```

```
                employee.getDepartment(),
```

```
                employee.getSalary());
```

```
        }
```

```
    } catch (SQLException e)
```

```

{
    System.out.println("Error displaying employees: " + e.getMessage());
}

}

public static void main(String[] args) {
    String url = "jdbc:mysql://localhost/ri_db";
    String username = "test";
    String password = "test123";

    try (Connection conn = DriverManager.getConnection(url, username,
password);
        Scanner scanner = new Scanner(System.in)) {

        int choice;
        do {
            choice = scanner.nextInt();

            switch (choice) {
                case 1 -> addEmployee(conn, scanner);
                case 2 -> updateSalary(conn, scanner);
                case 3 -> viewEmployeeRecord(conn, scanner);
                case 4 -> displayAllEmployees(conn);
                case 5 -> System.out.println("Exiting Employee Management
System.");
                default -> System.out.println("Invalid choice. Please try again.");
            }

        } while (choice != 5);

    } catch (SQLException e) {
        System.out.println("Database Error: " + e.getMessage());
    }
}
}

```

**Status : Correct**

**Marks : 10/10**